

Pre- Feasibility Study to Cultivate and Produce Dried Tomatoes



2017



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1. Executive summary

The project aims to cultivate and produce dry tomatoes of varieties suitable for drying and packing them in vacuumed bags with a weight of 0.5-1 kg and selling them in the local market, or capacity of 25 kg which targets the export markets in the European Union countries especially Eastern, Gulf countries and the markets of Russian Federation and Balkan countries.

Table 1: Economic Indicators

Project Name:	Cultivate and produce of dried tomato
Project Location	Ma'an
Project products	Dried tomato with a weight of 0.5-1-25 kg
Total Number of employees	27
Total investment (Thousand JD)	399
Net present value of cash flows (Thousand JD)	1,275
Pay-Back Period (year)	3
Internal Rate of Return (IRR)	33,3%

The idea of project is summarized by drying tomato through the reduction of water contents in the products and increasing the solid contents of the pulps by exposing tomatoes to sun for 7 days or more, this process will reduce the juice contents and increase the proportion of pulp, after cutting it to halves and add salt to get rid of moisture to concentrate only on the solid substance, and every 8-10 kg fresh tomatoes given a kilogram of dried tomatoes. It can be used in sauce or can be added to pizza or packaged in small packages after adding olive oil and other spices.

The project is expected to provide 27 new job opportunities , five indirect employees and twenty two direct employees, and the investment cost of this project is estimated 398,998 JD. This amount is determined through fixed assets, which include cost of the irrigation system, drilling of the groundwater wells, initial cultivation requirements of tomatoes, irrigation systems, machinery and equipment, furniture and transportation, pre-operating expenses and initial working capital. (50%) of the project will be financed through loans and through self-financing. Despite the high cost of investment for the project, it provides many jobs opportunities suit with the size of investment, because the project adopts on the employment in the process of harvesting, drying and selling the crop.

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The results of the financial analysis indicate that the net present value of cash flows for the project is positive, which is indicative of the financial feasibility of the project. The rate of internal return is much higher than the weighted average of capital cost, which is indicative of the feasibility of the project. Other financial indicators indicate that the project is financially profitable to the investor.

2. Key Highlight of Jordan

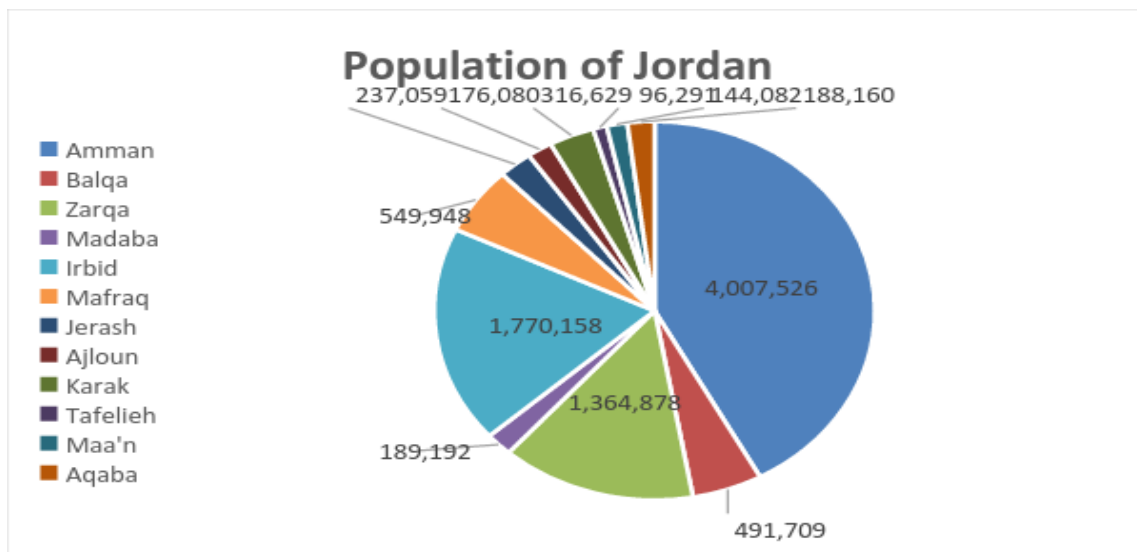
Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates. Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1: Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafrq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table 2: Population Distribution according to Age Group

Age Group	% Population	Males	Females
Years 14-0	35.04%	19.2%	18.2%
Years 64-15	61.02%	30.7%	28.6%
More than 65 Years	3.94%	1.6%	1.6%

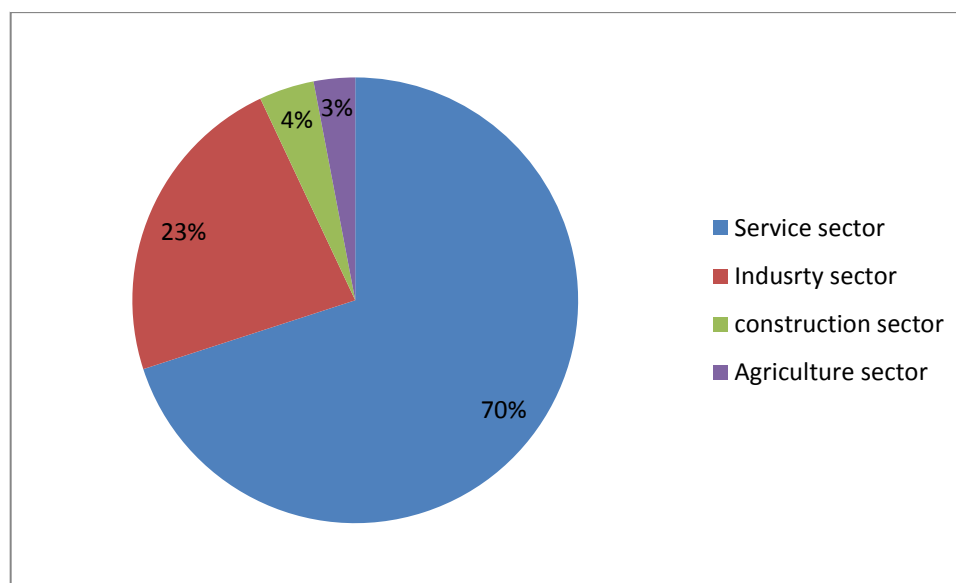
Department of Statistics (DOS), 2015

Overview of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure 2: Sectors Contribution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US\$3.8 trillion market and comprising 380 million consumers.

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Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally. The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce.

Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater

investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

Furthermore, in its endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Investment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

3. Market analysis

Project Description

The project is based on the idea of planting 200 donums of tomatoes using drip irrigation method, drying the tomatoes, packing them in appropriate bags that matching the export requirements. Therefore, the idea of the project is to create a special complex to prepare and dry tomato in sunlight by reducing its juicer and raise the concentration of solid content of tomato, to the extent that ensures the cessation of the activity and growth of bacteria and to stabilization of enzymatic activity that occurs inside the fruits, to preserve it for a long time while maintaining its chemical properties in a state close to its normal state without fermentation or damage.

Such projects are considered complementary and integrated agricultural projects linked directly to agricultural production and are often does not need high capital, its with commercial nature that require a large labor, where it depend on the labor to carry out the cultivation, harvesting, sorting, drying, packaging and handling.

Project Objectives

Due to the increase of dried tomato prices in the local market, which amounts to 18 JD per kilogram, there is a reluctance by consumers to buy this products because of the availability of substitute fresh tomato throughout the year, but sometimes at high prices. This item may be entered into the local market if the prices are reasonable. There are small quantities imported from EU countries, in addition to the limited quantities produced from small companies in Jordan ..

The project aims to:

1. Achieving a profitable return to the project owners.
2. Cultivation and production of tomatoes of suitable varieties for drying and supplying them to the local market with distinctive quality, and free of mold and moisture,
3. Create job opportunities in less developed areas, and changing the traditional agricultural production pattern of tomatoes cultivation
4. Export dried tomatoes and enter into competition with the world markets.
5. Increase domestic exports to Europe and other countries, which participate in the improvement of the national economy

6. Drying tomato is the best solution to overcome the variability and fluctuations of the market prices and ensure a stable return throughout the year.

Proposed Product

The solar drying method is one of the old ways that known, it was used to conserve the excess production for consumption either at times when the material is not available or to made it available off-season and use it at the time of need.

It can be said that solar drying is one of the oldest methods of food preservation known to the world population and known as the method of drying, human has resorted from ancient times to use and exploitation of solar energy, taking advantages of high temperatures in drying the excess production of various foods in order to save it through possible longest period of time and use them before the next season. Also, people invented many ways to save vegetables such as tomatoes, okra and Jews mallow and others, in addition to fruits such as grape, dates, figs and many fruits. Now adays, there is modern methods are used for drying such as: industrial methods that use industrial dryers to control the speed of air passage, relative temperature and humidity inside the dryer and a low temperature preservation, canned food and others .

The idea is to produce dried tomato through the cultivation of tomato crop of suitable varieties for drying, then the tomato is dried using the solar radiation method, and after the completion, tomato is packaged in a good manner and then exported or distributed to local markets in Mega Malls, shops, restaurants or to export markets abroad.

Target Segments

- Export markets (Eastern Europe in particular, Italy and other EU countries).
- Arab markets and the markets of Gulf countries in particular
- Local market (restaurants serving Western meals and families)

Market Size Analysis

The data of the Department of Statistics indicate that the average cultivated area with tomato amounted at about 136 thousand donums during the period (2011-2015) with an estimated production capacity of 800 thousand tons (Table 3). And that the production of these areas of tomatoes meet the full domestic demand estimated at about 40% of the production, also tomatoes are exported to the Arab markets, mainly the Gulf markets.

It is also noted that the unplanned expansion of tomato cultivation has caused an increase in crop exceeds the needs of the local market, leading to the establishment of some agro-industries (such as Tomato paste) to solve marketing problems, not for manufacturing purposes.

Table 3: Development of cultivated area, production and productivity of tomato in Jordan

Year	Cultivated Area (dunum)	Production (Ton)	Productivity (ton/dunum	Total irrigated area in Jordan (Dunums)	Percentage of cultivated area with tomatoes to the total irrigated area
2011	129,536	777,820	6.005	869,138	%14.90
2012	123,445	738,227	5.980	869,138	%14.20
2013	154,338	869,138	5.631	869,138	%17.76
2014	145,641	744,602	5.113	869,138	%16.76
2015	128,871	870,017	6.751	869,138	%14.83
Average	136,366	799,961	5.896	869,138	%15.69

Table (4) shows the value and quantity of tomato exports during the period (2011-2015). The average quantity of exports of fresh tomatoes reached 480 thousand tons valued at about 216 million JD.

Available data indicate that exports of fresh tomatoes to Arab countries and the Gulf States are still modest to entry the markets with high-income levels. The data also indicate a significant moderation in the value of exports, with an annual value of about 206 million JD during the period (2011-2015).

According to the statistics of foreign trade, the average price of FOB per kilo of tomatoes exported to the Gulf markets was 450 fils. These prices reflect the apparent decline in the value of one unit of exports, which can indicate the lowquality of Jordanian exports to these markets compared with those coming from competing countries.

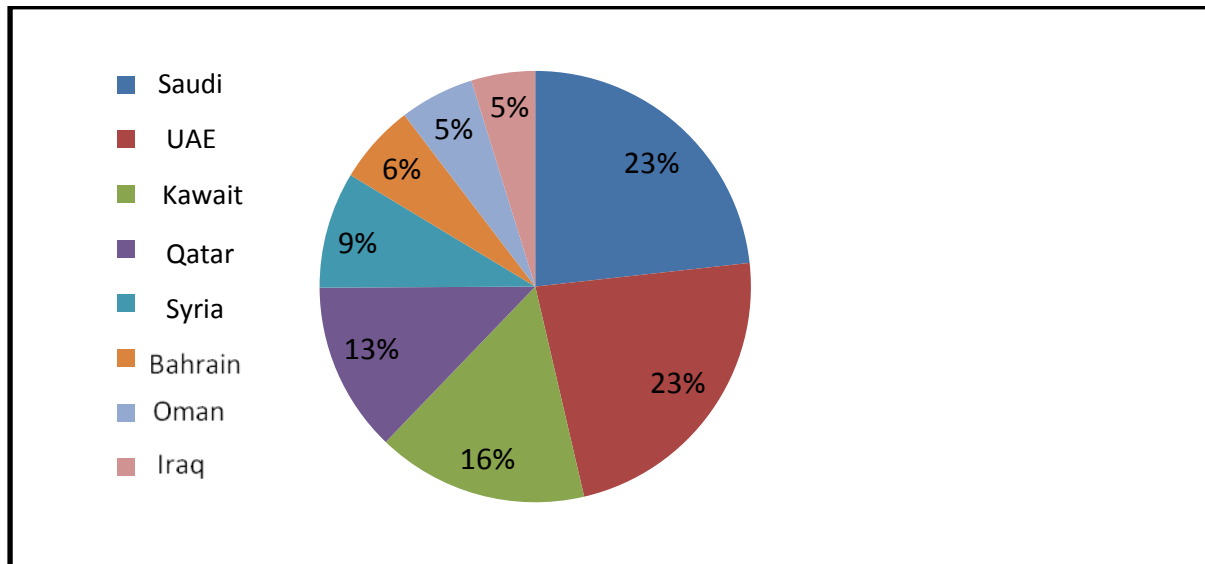
Table 4: Value and quantity of exports and trade surplus of fresh tomatoes (70200000)

Year	Value of exports (1,000 Jordanian Dinars)	Quantity of exports (Ton)	Price per ton of exports (JD)	Trade surplus (1,000 Jordanian Dinars)
2011	159,637	434,806	367	159,480

2012	177,636	418,475	424	177,645
2013	224,587	611,519	367	224,587
2014	283,293	517,207	548	283,293
2015	235,427	419,271	562	235,431
Average	216,116	480,256	454	216,087

Source: Department of Statistics (2015) Foreign Trade Bulletin.

Figure 3: Fresh tomato markets exported from Jordan



Available data indicate that horticultural exports to EU countries are still modest in terms of the entry to high-income level markets , so that the amount of exports to non-traditional markets remains modest and has not exceeded the 3,245 tonnes per year (Table 5) despite the opening of theses markets to Jordanian horticultural products as a result of the signing of the Association Agreement with European Union in 1998. It is believed that the main reason for this problem is the inability of national marketing projects and the agricultural marketing system to deliver Jordanian products to export markets in a competitive and qualitative manner.

Table 5: Jordan's exports of fresh tomatoes to EU countries during the period (2011-2015)

Fresh Tomato	Value of exports (thousand euros)					Quantity of exports (tons)				
	2011	2012	2013	2014	2015	2011	2012	2013	2014	2015
Years										

Bulgaria	657	257	15			1,316	530	32		
Croatia			8					9		
Czech Republic	18					20				
France	11					8				
Germany	0									
Greece	72	1				79	1			
Hungary	181	2				201	4			
Denmark	95	0	8			102		8		
Poland	87					74				
Romania	2,287	476	157	51	15	11,052	2,352	383	59	38
Sweden		0	0							
United Kingdom		1					2			
European Union	3,407	737	189	51	15	12,852	2,889	432	59	38

Source: EU Commission (2017)

Dried Tomato have the number (HS 07129030) is in the International Harmonized System (HS). There are no data on the quantity of imports or exports of this commodity in the data of department of statistics in Jordan , which may be incorporated either with fresh tomatoes (0702), Or preserved tomatoes as shown in **Table 6**. In a review of the EU Export and Import Database shows that small amounts of dried tomatoes were imported from Italy at an average price of € 4.8 per kilogram. **Table 7** shows Jordan's imports of dried tomatoes from EU countries.

Table6 : HS code for different types of Tomato

HS Code	Commodity_Description
0702	Tomatoes, fresh or chilled.
071290300	Vegetables, , Dried And Vegetable Mixtures, Dried (Whole, Cut, Sliced, Etc.), But Not Further Prepared
200210000	Tomatoes , whole or in pieces , prepared or preserved other wise than by vinegar or acetic acid .
200290100	Tomatoes , prepared or preserved otherwise than by vinegar or acetic acid, other than whole or sliced, in packings for retail sale not exceeding a net weight of 3 kg.
200290900	Tomatoes, prepared or preserved otherwise than by vinegar or acetic acid, other

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HS Code	Commodity_Description
	than whole or sliced, in packings for retail sale with a net weight more than 3 kg
200290000	Tomatoes, prepared or preserved otherwise than by vinegar or acetic acid

Table 7: Jordan imports of dried tomatoes from EU countries during the period (2011-2015)

Dried Tomato	Imports value (euro)					Imports Quantity (Kg)				
Years	2011	2012	2013	2014	2015	2011	2012	2013	2014	2015
Germany		943	415				189	83		
Italia	27,992	13,441		1,575	9,593	9,000	1,000		350	2,000
Holanda	386									
Spain					403					
European Union	28,378	14,384	415	1,575	9,996	9,000	1,000	83	350	2,000

(Table 8 and Table 9), show the volume of EU imports of dried tomatoes, indicates that Turkey, Tunisia, China, Israel, and Morocco are among the most important exporters of dried tomatoes to EU countries outside the EU group with a quantity of 16,000 tons per year. The intra-OIC trade is estimated at 31,000 tons of dried tomatoes amount to 93 million euros. Table 9 indicates that the price per kilogram of dried tomatoes exported to the EU countries ranges between (2.3 - 7.1) Euros based on source and quality. Table (10) indicates the quantity of dried tomatoes exported from the EU to some Arab countries. However, the annual average is estimated to 120 tons of dried tomatoes, where arab countries import dried tomatoes from Indian continent markets.

Table 8: EU imports of dried tomatoes from outside the EU during the period (2011-2015)

Years	2011	2012	2013	2014	2015	Average
Turkey	6,550	6,171	7,787	8,633	9,009	7,630
Tunisia	5,255	5,745	5,038	5,570	5,858	5,493
China	2,686	2,690	2,560	2,411	2,265	2,522
Israel	592	458	349	235	354	398
Morocco	127	1	170	101	247	129

Egypt	432	400	145	40	111	226
Chile	0	0	0	62	108	34
United State	0	3	18	14	34	14
Albania	0	0	0	1	16	3
Swaziland	128	91	41	11	14	57
Uzbekistan	2	5	4	6	5	4
Senegal	2	158	45	16	3	45
European Unio	15,821	15,807	16,247	17,131	18,029	16,607

Table 9: Value and quantity of EU imports of dried tomatoes from EU countries during 2015

Dried Tomato HS (07129030)	Import value (000 euro)	Import quantity (Ton)	Import price (euro/ Kg)
Turkey	27,968	9,009	3.104
Tunisia	13,567	5,858	2.316
China	7,723	2,265	3.410
Israel	1,267	354	3.580
Morocco	894	247	3.621
Egypt	730	108	6.764
Chile	265	111	2.390
United State	242	34	7.119
Albania	94	14	6.744
Swaziland	21	5	4.109
Uzbekistan	52,827	18,029	2.930

Table 10: Quantity of EU exports of dried tomatoes per tonne to some Arab countries

Year	2011	2012	2013	2014	2015	المتوسط
Bahrain	2	1	5	10	9	5
Egypt	57	50	32	63	49	50
Jordan	9	1	0	0	2	2
Kuwait	8	4	4	19	17	10
Lebanon	1	2	3	4	0	2
Libya	0	0	0	0	1	0

Amman	1	2	1	1	3	2
Qatar	2	15	4	3	17	8
Saudi Arabia	5	4	6	31	30	15
United Arab Emirates	23	17	32	23	24	24
Arab countries	108	96	88	154	152	120
Outside the EU	1,757	1,525	1,747	1,666	1,449	1,629

It is difficult to estimate the accurate amount of local demand for dried tomatoes due to the exaggerated figures, and local high prices, and the market survey shows that the average price per 1 kilogram of dried tomatoes amounts to 18 JD per kilogram, this high price makes the elasticity of demand for the product up to zero, as a result of Fresh, canned and preserved alternatives.

However, if the price falls to (2.5-3) JD per consumer, it is expected to increase the demand for dried tomatoes to be used in pastries, pizza, Tajine and salads, and stimulate the demand for dried tomatoes by restaurants, especially at times when the price of tomatoes increase during the periods where there is no tomato production in valleys, and tomatoes in high areas have not yet mature.

Based on this, 50 tons of production will be directed to the local market, 100 tons to the Arab countries markets and 100 tons to the markets of European countries and Eastern European countries. Therefore, the capacity of project was estimated at 255 tons per year of dried tomatoes, which can be produced through a single 200 dunum farm or 100 plastic houses in protected agriculture. It is possible to contract with exporters, hypermarkets, supermarkets, hotels, villages, resorts, tourist restaurants and other restaurants. This reduces the daily demand for fresh tomatoes and the cost of conserving and storing them.

Competitors Analysis

The local market needs of dried tomatoes is covered by the quantities that are dried and stored at the household level. There is no factory in Jordan that is drying tomatoes except Rum Company, which started several years ago with the production of dried tomatoes, Rum is drying tomatoes that do not conform to the marketing specifications for fresh consumption due to the small size of fruits and relying on excess production at the time of lower prices. The estimated quantities of production of Rum company about 30 tons per year are marketed locally and some of the production is exported to some Arab markets with an average price of about 2 JD/kg.

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Therefore, the project is expected to face difficulties in achieving the expected sales if it targets the local market only at high prices as currently prevailing. Therefore, the investor must develop a good marketing plan and employ competencies with the ability to communicate with dried tomato importers in the Arab markets and European markets in addition to the emphasis on the role of advertising of the project at the beginning of its work to introduce the product, which enhances the opportunities of local marketing.

Expected Occupancy Rates

The importance of drying tomato is to keep the fruits from being damaged for as long as possible. This product is also very needed in the European markets. It is used in the preparation of pizzas, pasta and pastries, as well as home meals such as Tajen and others. In addition to the demand in supermarkets, hypermarkets in different regions, regular restaurants and other tourist and major local hotels, but due to the increase of prices, which is not justified and irrational, the demand is very low currently because of the availability of alternatives for fresh tomatoes and preserved tomatoes in iron cans, but the demand can rise if the prices is reasonable.

It is possible to calculate the net volume of the previous domestic demand based on the calculation of the quantities produced locally that estimated at 30 tons from Rum agricultural company and can similarly estimated the quantity produced in household levels in about 30 tons, and the total local production will be 60 tons / year. In addition to the quantities that imported from the countries of the EU with an average of 2 tons per year. Therefore, the domestic demand is about 62 tons at the current high prices.

Assuming an increase in the demand for this type of products by 2% per annum, which is lower than the annual growth rate of the population, and conservative for the growth rate of fresh tomato exports of 6.6%. Demand for this type of product is expected to increase if external markets are opened, especially in countries where tomatoes are not available all the year, such as European countries, or the prices are higher in certain seasons of the year.

The project's market share can be estimated at about 225 tons per year of the cultivation of 200 dunums in the desert lands in Ma'an based on the estimated productivity of the project amounted at (8.5) tons / year of fresh tomatoes and converted to dried tomatoes by solar drying which is 15% of Fresh material weight. Production capacity in these projects depends on manpower and their skills rather than equipment . It was assumed that the project would start with half ton of fresh tomatoes per day and that it would work 150 working days per year, with a 2.5% annual increase in production.

It is worth mentioning that one kg of fresh tomatoes gives 150-200 g of dried tomatoes, and it is estimated that the production of the project in the first year of operation will amount to 128 tons , which constitutes about 21% of the expected market size for 2018. While, the fifth year it will represent about 21.4% of the

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expected market share for the same year. The future demand for tomato was estimated during the period (2018-2027) as shown table 11.

Price Analysis and Pricing Policy

Dried tomatoes are sold in some supermarkets and hypermarkets in small containers of 200-250 gm at prices ranging between 3-4 JD per container, as well as selling it at vegetable and dried fruits shops in an open bags exposed to dust at prices ranging between (8) JD for the local product and (18) JD per kg for the imported product from Italy. "Al-Basita agricultural industries" company in Jerash produce dried tomatoes, by adding oil and herbs and keeping it in glass containers of 250-500mm.

Based on the analysis of the prices of imported dried tomatoes and exported to EU countries, at the rate of about € 3 per kilogram from outside the EU. It is suggested that the price of 1 kilogram of dried tomatoes is (2) JD per kilogram, its the farm price without transporation, to give a profit margin for the wholesaler and retailer to sell it to the consumer at a price ranging from (2.5-3) JD depending on the quality of the varieties and their content of moisture. It is proposed that the average wholesale price of dried tomatoes throughout the year is about 2 JD / kg. The products can be marketed to major shopping malls, supermarkets and grocery stores.

Expected Revenues

The tomatoes are harvested during a period ranges between 6-7 months a year, so most farmers in Jordan choose tomatoes as a kind of risk management strategy, whereas the length of the production season ensures that they get very high prices at some times and very low prices at peak production time . The rate of productivity of one dunum of tomatoes in open desert highlands is between (8-12) tons per dunam. Table (12) shows the expected revenues from the project. For the purposes of this study, it was considered that the average sale price is 2000 dinars / ton on the farm price without transportation after the process of drying and conserve it in vacuum bags also it is expected to increase prices by 2% per year.

Table 11: Project share of the local market during the period (2018-2027)

Demand Indicators on dried tomatoes	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Expected population (1,000 people)	9,984	10,204	10,428	10,658	10,892	11,132	11,377	11,627	11,883	12,144
Annual consumption per capita (kg / person)	0.050	0.050	0.050	0.050	0.050	0.050	0.050	0.050	0.050	0.050
Total local consumption of the commodity	499,21	510,19	521,41	532,88	544,61	556,59	568,83	581,35	594,14	607,21
	1	4	8	9	3	4	9	4	4	5
Local consumption of goods per ton	499	510	521	533	545	557	569	581	594	607
Restaurant Request (Tons)	500	511	522	534	545	557	570	582	595	608
Local demand for goods per ton	999	1,021	1,044	1,067	1,090	1,114	1,139	1,164	1,189	1,215
Expected Project Production (Ton)	113	225	225	225	225	225	225	225	225	225
Project market share of local demand (%)	11.26	22.03	21.56	21.09	20.64	20.20	19.76	19.34	18.92	18.51

Table 12: Selling prices and expected revenues during the period (2018-2027)

Project income	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total sales (tons)	144	288	288	288	288	288	288	288	288	288
Item Price (JD / ton)	2,000	2,040	2,081	2,122	2,165	2,208	2,252	2,297	2,343	2,390
Total sales (JD)	288,000	587,520	599,270	611,256	623,481	635,951	648,670	661,643	674,876	688,373

4. Technical analysis

Firstly: Cultivation of tomatoes

English name: Tomato

Scientific name: *Lycopersicon esculentum*

Family: Solanacea

Native home: Central America and South America

Origins of Varieties: Most spherical commercial varieties follow plant variety.



Plant Description: Tomato is annual herbaceous plant. **Roots:** The plant has a root deep in the soil in case of seed planting directly in the permanent field its rarely used). But, in the case of seedlings planting, the primary root is often cut when the seedlings are harvested from the nursery and instead grows a dense root total of 15 20 lateral roots. As the plant progresses in age, the ability of the root to absorb is reduced and its activity gradually decreases, leading to gradual plant death after the end of the harvest season. The stem of the tomato plant is round in the cross section and is covered with thick hairs. It grows vertically at the beginning of growth until it reaches 30-60 cm and becomes bending in the unlimited varieties. The leaves of the tomatoes are composed of feathers and consist of 7-9 opposite plates (depending on the variety). They are grown with small leaves and the neck is long. The leaves are seated and their edge is divided into most varieties. It is covered with thick hairs and has a distinctive odor that appears in the early morning or when pressed with the fingers This smell is characteristic of tomato leaf about potatoes.

Botanically, a tomato is a fruit, a berry, consisting of the ovary, together with its seeds, of a flowering plant. Their flowers, appearing on the apical meristem, have the anthers fused along the edges, forming a column surrounding the pistil's style. Flowers in domestic cultivars can be self-fertilizing. The flowers are 1–2 cm across, yellow, with five pointed lobes on the corolla; they are borne in a cyme of three to 12 together

The categories are divided according to the production methods and the purpose of agriculture:

A) Fresh Market

B) Processing

C) Home gardens

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D) Protected cropping

C) Mechanical Harvesting

H) Drying varieties

And there are many dryable varieties grown in Egypt, including (Red Spring, Taylor, Angos) The most popular dryable varieties are:

- Carol Chyko Tomato
- Costoluto Genovese Tomato
- Coyoto Tomato
- Federle Tomato
- Opalka Tomato
- Principe Borghese Tomato
- Russo Sicilian Toggeta Tomato
- San Marzano Redorta Tomato
- Zapotec Tomato

Conditions for selecting good items

1. Solid Salad tomato varieties with empty compartments that are not suitable for good dried tomatoes.
2. Do not prefer varieties that have aliases or external protrusions on the fruit. Fruits must be smooth.
3. Long round and long-standing fruits are suitable for drying, but long-standing varieties are best for obtaining dried tomatoes. (One of the most famous varieties in China is the long-standing banana tomatoes. In Tunisia, Sabra from the production of american petuseed is one of the best dried varieties.
4. Medium-sized fruits varieties are best for drying and are required for export.
5. The product should be characterized by the homogeneity of the fruit sizes. The variation in the final volume of the dried fruit reduces its degree and thus its price.
6. Varieties with an amber neck are not desirable in dried fruits.
7. The variety should be characterized by the thickness of the outer layer, which is an important export feature. If the outer layer in the final product is thicker, the grade and the price will increase.

The fruits after drying are treated to market to the final consumer in several processing to recover part of its normal size and the greater the thickness of outer layer, the higher the final volume obtained.

8. Do not prefer high fruits in the degree of pH
9. Large sized fruit with dense juicer is not suitable for drying, as it takes a long time in drying and it exposed to molds in the racks, and these varieties are preferred in the manufacture of ketchup.
10. The required color grades are full red in their degrees and dark orange and do not accept orange or light orange fruits.
11. In general, the product should be characterized by high content of total soluble salts (TSS) and dry matter.

Conditions to be provided in tomato drying areas:

1. The area has heat and sunny atmosphere.
2. Far from the areas and seasons of rainfall where rainfall leads to rotting fruits above racks.
3. The place should be far from the wind, particularly the dusty places. It is best to be protected by windbreaks, as the strong wind may fall down the racks on the ground, causing the loss of the crop due to contamination.
4. Availability of employment.
5. Provide the necessary places for the process of packing tomato after drying, as well as warehouse for cold storage.
6. The site of racks should be close to the areas of agriculture and harvesting.
7. Racks should be higher than the soil level and can be wire not rusty.
8. It is preferable to cover the racks with green nets to protect them from dust.
9. It is preferable to be racks with high edges to prevent the falling of fruits on the ground.
10. Ideal ambient temperature for drying is 20-35 °C, which should be available when drying the fruits by racks.

Required Raw Materials

There are many agricultural operations required to produce tomatoes starting from land preparation, tillage, settlement, cultivation, fertilization, pesticide spraying, and irrigation, in addition to harvesting, transferring, cutting, drying, batching, rolling, packing and stamping the crop in small bags or containers.

Among the materials needed for the production of tomatoes are seedling, organic fertilizer, chemical fertilizer of various types, pesticides, resistance to weeds and irrigation water. It is known that tomato consumes medium amounts of water compared to other agricultural crops due to its permanence in the

soil, the density of vegetation and the growth speed . The average water needs of dunums are around 600-700 cubic meters per year. Table 13 shows the cost of raw materials for agriculture and the annual cost of cultivating 200 dunums of tomatoes under drip irrigation system

Table 13: Prices, quantities and cost of annual raw materials required for agriculture

Production Requirment	Unit	Quantity per donum	Annual quantity (Unit)	Value (JD per dunam)
Seeds	Seedlings	0.06	2000	120
Domestic Fertilizers	Ton	20	2.5	50
Chemical fertilizers	Kg	0.5	320	160
Waters	M ³	0.21	600	126
Mulch	Meter	0.5	40	20
Pesticides	Leter	17	10	170
Others	JD/ Donum	1	5	5
Salt and chemicals	JD/ Donum	0.5	20	10
Carton Packages	JD/ container	0.15	800	120
Glass jars	JD/ container	0.1	200	20
Plastic Packaging	JD/ container	0.05	800	40
Bags				
Total				841

Required Human Resources

This project requires an agricultural engineer of plant production specialized in tomato production, in addition to food production engineer to ensure high quality, agricultural and food production technician, as well as an employees who resbosable for marketing product, accounting, bookkeeper, agricultural workers and tractor driver. The following table (Table 14) shows the employment requirements for the project:

Table 14: employment requirements for the project

Job Title	Number
Project Manager / Agricultural Engineer	1
Agricultural Engineer	1
Food Production Technician	1

Marketing and supply representative	1
Guard	1
Total indirect employees	5
Technical production supervisor	1
Workers	20
Driver	1
Total direct employees	22
Total number of employees	27

Administrative costs Include: the salaries of executive director, an agricultural engineer /food production, supervisor and production technician , marketing and supply of raw material and guard, etc .. (Table 15 and Table 16), to reach the cost that associated with the administrative costs to (125.400) JD per year,the direct employment salaries increased when the production capacity increased up to 2% per annum with an increase in employment experience. Furthermore, the health insurance for workers, the contribution of social security allowances and transportation allowances has been added, so the annual salary calculated by 16 months instead of 12 months to cover these costs and to estimate the total annual labor wages, including the indirect employment wages as shown in (Table 17)

Table 15: Direct and Indirect Employees

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect Employees										
Project Manager	1	1	1	1	1	1	1	1	1	1
Agricultural Engineer	1	1	1	1	1	1	1	1	1	1
Food Production Technician	1	1	1	1	1	1	1	1	1	1
Marketing and supply representative	1	1	1	1	1	1	1	1	1	1
Guard	1	1	1	1	1	1	1	1	1	1
Total number of indirect employees	5	5	5	5	5	5	5	5	5	5
Direct Employees										
Technical production supervisor	1	1	1	1	1	1	1	1	1	1
Workers	20	20	20	20	20	20	20	20	20	20
Driver	1	1	1	1	1	1	1	1	1	1
Total number of direct employees	22	22	22	22	22	22	22	22	22	22
Total	27	27	27	27	27	27	27	27	27	27

Table 16: Monthly salaries for direct and indirect employment

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect Employees										
Project Manager	1,000	1,030	1,061	1,093	1,126	1,159	1,194	1,230	1,267	1,305
Agricultural Engineer	800	824	849	874	900	927	955	984	1,013	1,044

Food Production Technician	400	412	424	437	450	464	478	492	507	522
Marketing and supply representative	350	361	371	382	394	406	418	430	443	457
Guard	300	309	318	328	338	348	358	369	380	391
Direct Employees										
Technical production supervisor	350	361	371	382	394	406	418	430	443	457
Workers	300	309	318	328	338	348	358	369	380	391
Driver	300	309	318	328	338	348	358	369	380	391

Table 17: Salaries and wages of direct and indirect labor

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect Employees										
Project Manager	16,000	16,480	16,974	17,484	18,008	18,548	19,105	19,678	20,268	20,876
Agricultural Engineer	12,800	13,184	13,580	13,987	14,407	14,839	15,284	15,742	16,215	16,701
Food Production Technician	6,400	6,592	6,790	6,993	7,203	7,419	7,642	7,871	8,107	8,351
Marketing and supply representative	5,600	5,768	5,941	6,119	6,303	6,492	6,687	6,887	7,094	7,307
Guard	4,800	4,944	5,092	5,245	5,402	5,565	5,731	5,903	6,080	6,263
Total number of indirect employees	45,600	46,968	48,377	49,828	51,323	52,863	54,449	56,082	57,765	59,498
Direct Employees										
Technical production supervisor	5,600	5,768	5,941	6,119	6,303	6,492	6,687	6,887	7,094	7,307

Workers	96,000	98,880	101,846	104,902	108,049	111,290	114,629	118,068	121,610	125,258
Driver	4,800	4,944	5,092	5,245	5,402	5,565	5,731	5,903	6,080	6,263
Total number of direct employees	106,400	109,592	112,880	116,266	119,754	123,347	127,047	130,859	134,784	138,828
Total	152,000	156,560	161,257	166,095	171,077	176,210	181,496	186,941	192,549	198,326

Job Description:

- The agricultural engineer is fully aware of the varieties of tomatoes suitable for drying, how to obtain them, when to grow them, and the ability to request and provide the necessary materials from the plant production tools and supplies, processing the suitable agricultural land for growing the crop, Irrigation scheduling and implementation of prevention and pest control measures in accordance with environmental standards. Harvesting and harvesting of products, assembly and drying of products, treatment of products, drying and pressing, and supervision of crop harvesting, drying and transport. As well as the mobilization of farm records and work models, documenting expenditures and preventive control measures and their timing, proposing policies and action plans, monitoring their implementation and evaluating their results. Organizing the daily work, distributing the employees, following up the implementation of their work, and coordinating with the different production parties to accomplish the required tasks as planned. Determining the varieties of tomatoes suitable for agriculture, identifying the types of pesticides needed for different types of plants, their dates and quantities, in addition to determining and securing the quantities of fertilizers required in coordination with the technical specialists. Coordinate with the transport companies to ensure the safety of the production process and to achieve the appropriate return, verify the accuracy of the quantities delivered and received, and organize the statements and records of their own. Check the availability of requirements and requirements for the safety and security of stored materials from pesticides, natural and chemical fertilizers and do not harm the environment and public health, to make proposals aimed at improving the quality and quality of production. Supervising the technical and management of the staff and labors, evaluating their performance and achievements and proposing programs to raise their knowledge and develop their abilities. Develop procedures and ensure safety and occupational health and public facilities.
- Food Industrial Engineer: good knowledge of drying process, tomatoes varieties suitable for drying, preparation the specification of products, determine the degree of humidity and degree of drying, how to conduct the production process, identification of the raw materials and the appropriate time schedule for its supply, follow drying procedures of the crop. Preparation of the production plan according to the degree of the product required to develop the appropriate

technical solutions for the problems related to the production process or the quality of the product. Identification of the raw materials involved in the composition of the product, in terms of type and percentage of use and compliance, and determining the appropriate temperature and humidity for the storage of raw materials and food products and their validity, Matching the results of lab tests of the samples drawn with quality specifications, and developing procedures and methods of work and production using modern technologies. Development of quality systems and standards. Prepare technical reports on food processing operations and implement procedures and ensure occupational and public health and safety requirements

- **Agricultural Technician:** Preparing the land for growing crops by tillage, laying and softening, installing irrigation system apply chemical and organic fertilizers. Provide materials and work equipment of seeds, seedlings and pesticides according to the quantities and specifications specified. adding organic fertilizers to land and cultivation of seed crops by species and varieties, preparation and elaboration of raw materials specifications and the appropriate time schedule for their time of application. Monitoring the procedures and maintain the machines and preparing the appropriate operating programs for them, preparing programs for the maintenance of machines and equipment, identifying the materials needed for the production plan and providing them in the warehouses with their quantities and timetables for their use, and developing appropriate technical solutions for the problems related to the production process. Preparing technical reports on agricultural and production operations, and implementing procedures and securing occupational and public health and safety requirements.
- **Marketing Officer:** preparing for the receipt of the product, testing and matching the quantities, types and specifications with the requested order, inspecting its validity for human consumption, filling the forms of the receipt and delivery process, following up the loading and unloading process, negotiating the selling prices, explaining the advantages and specifications of different types of products, analysis the customers needs, response to their inquiries, the securing of their requests, the follow-up of the processing, the weight of the materials, the sale of the product, issues of bills and receive the payments, the editing of the invoice, the inventory of the contents of the stores, Lists of shortages of raw materials, ensure sufficient storage of raw materials and final products. Organization of sales records and financial records.

Mobilization of business models. Application of occupational safety and health procedures and instructions.

- Agricultural worker (drying, packing and packaging): Working under the supervision of the agricultural engineer. conduct the industrial process as requested, use of drying and packaging tools, receipt of the crop from the field, washing and cutting and transfer to the drying field, adding salt, sorting the product processing machine, packing and packaging of the products and store the products until marketing. clean the packaging equipment and containers. Place the packaging on the production line, sorting the product and removing the distorted and damaged parts
- Agricultural worker: works alone or under the supervision of the agricultural engineer and prepares the necessary materials. Preparing the land for cultivating by plowing, clearing, preparing and preparing the materials to used in the agricultural land, planting the seeds, take caring of the crops, monitor germination, flowering, spraying the plants with pest control materials such as insecticides, control the diseases and rodents in accordance with environmental standards. remove infected parts, protect the crop from the risk of frost. Picking and harvesting products, packaged, stored it in the storage rooms. follow the occupational safety and health procedures and instructions.

Location Analysis

Tomato plants are plants that bear various environmental conditions. It tolerate low to frost temperatures, as well as high temperatures in the summer. Also, it tolerate salinity of soil and water. The higher the quantity of water and fertilizers given to tomatoes, the greater the productivity. Therefore it preferable to cultivate it in Ma'an.

When selecting a project site, the following should be considered:

- Ease marketing and transportation of raw materials and its products for both internal and external marketing.
- Provide a permanent drinking water source that is clean and free of permanent minerals.
- Easy disposal of solid and liquid waste, which is an important health condition in the factory.
- Ma'an region and the desert and eastern areas are one of the suitable sites for this factory because it provides sufficient land, intensity of solar radiation and lack of rainfall.

In the proposed area, all necessary services are available from paved roads, water and electricity. Electricity can be supplied from the electricity company as well as water from wells near the farm in case of need.

Production Process Description

It is suggested to use solar drying, which is low in cost, this method common in warm countries where sunlight is available most of the year, such as the countries of the Mediterranean basin, Africa, California and Australia. Tomato drying steps are as Follows:

1. Wash fruits to get rid of dirt, bacteria and residues of agricultural pesticides.
2. Manual sorting of the fruit that is not mature or damaged or that is not suitable for drying to provide defects
3. Cut the tomatoes into slices according to the demand.
4. Spread the slices on the drying trays, and saline the sliced salt with 5% of the weight of the fruit to accelerate the drying process and keep the material from being damaged, placed on trays and dried outdoors for 4-7 days in warm areas.
5. Dried fruits are collected and sorted into the grades (Extra, First Class or Second Class)
6. The product is wrapped according to the requirements of the target market.

Drying method

Drying method

1. Red ripe fruits are collected , provided that they are not excessive in their maturity, ie, excess redness, where the dark fruits are transformed after drying to brown color. This is rejected by exporting. The color must be complete red or dark red (without increasing maturation) or dark orange. Yellow or light orange or green colors are not accepted.
2. Then the fruits are washed, sorted and Exclude the fruits of the incomplete color, green, yellow, and injured by mechanical injuries or molds
3. The process of cutting the fruit longitudinally to halves are identical exactly without full separation of the two halves and keep the crust between the halves so as not to tend dried tomatoes on the drying racks and prolong the period of drought.
4. It should be cut on high racks for soil so as not to pollute fruits, and the rate of completion of one worker in the process of cutting from 300-500 kg / day / worker.

5. In some countries, such as the United States, fruits are treated after being cut with sulfur dioxide for disinfecting the dishes, but this gas is not allowed in EU exports.
6. After cutting, it salted with a concentration of no more than 30% or sprayed by hand, which is preferable, as the solution has sometimes mold some slices.
7. The fruit is then separated on clean racks, so that it is Compacted without leaving gaps between the fruits. The racks was above the surface level so its not contaminated.
8. It takes time to get good drying fruits (which is not less than 10% humidity and not more than 20% over 3-12 days depending on the area).

Notes:

1. The final net ratio is 1 kg dry tomatoes obtained from 10-12 kg fresh tomatoes.
2. After separating the cutting tomatoes on the racks and salted, they are checked after 3-4 days to follow the drought rate so as not to dry the fruits too much and therefore lose a large part of the crop.
3. After the final product reaches the desired drying rate of 10-20%, it is collected in clean areas to avoid contamination and then be scaled according to their size. The prices increase to the larger sizes and less to the lower size. Then, they are packed in containers weighing 10-15 kg.
4. During sorting and packaging, samples of dry fruits are taken to analyze pesticide residues and measure some other quality indicators such as acidity and TSS.
5. This followed by storing in refrigerators until shipping at 5-10 ° C and for up to one year.



Dual slicing system for tomato drying

Drying the tomato will reduce the moisture of the tomato to be dried until the moisture content in dry matter reaches about 10%. The dried tomato should be fresh, full of maturity and healthy, it is preferably drying the tomato on the same day of harvesting, because the fruits of mature tomatoes are perishable because of the high content of water

Direct dryers can be made directly using trays made of metal mesh loaded with stone or wooden blocks to allow air to pass under the crop. A layer of gauze cloth is applied over the crop to protect it from insects and birds during drying. Examine the crop daily, taking it to a closed place or under a roof if we expect rain.

An easy way to solar drying is to create a high frame of wood and cover this frame with a mesh and the use of straw mats. The air passes over and under the crop, which increases the speed of drying with the reduction of losses caused by excess heat

The trays, mesh or mats used to dry the crop must be clean. trays made of stainless steel, plastic or nylon are much easier to clean than wood. To reduce the chances of mold growth on trays, mesh and drying mats when not in use, they are well washed, dried and stored in a well ventilated area

During harvest operations, workers in the field can carry the infection to fruits or vegetables by touching them with dirty hands or knives. toilets or handwashing facilities must be provided and should be used by harvest team members. Monitoring, following up, evaluating and implementing cleaning instructions such as personal hygiene and hand washing after use of the toilet is necessary to reduce the infection of human pathogens. Those who have symptoms of dizziness or hepatitis A, who are sick with hepatitis C virus or diarrhea, should not be charged with collecting fresh crops.

Technical Requirements Analysis

The most important technical requirements for tomato cultivation project which using drip irrigation systems can be summarized as follows:

- A flat land for the cultivation of tomato crop.
- Install drip irrigation systems with all their equipment and pumps.
- Installation of black Plastic bags before planting on drip irrigation pipes
- Cultivate tomato seedlings according to the desired and dryable types.
- Fixed source of water from groundwater wells.
- Drying equipment, trays and a drying rack to dry tomatoes.
- Electrical source or generators at the project site
- Tractors and agricultural equipment.

- Hangars, warehouses, administrative offices and housing.

Land

The project needs a total area of 205 dunums, about 200 dunums for tomatoes cultivation, and 3-4 dunums for services, facilities and corridors, housing for workers, as well as a werehouse for storing fertilizers, seeds, production requirements, Hangers for machines and agriculture equipments. It is proposed to rent the land from the agricultural lands in Ma'an . It is expected that the annual rental fee of the dunum will be no more than 40-50 JD per dunam. Thus, the annual land tax will be about 8,000 JD per year and an annual increase of 2%.

Construction works

The estimated project needs for the buildings are 570 square meters, which will provide the production hall, warehouses, management and workers services. It is preferred to be cemented with a suitable height that provides the required ventilation and meets the sanitary conditions of food industry from walls and floors .The project needs temporary housing for workers, hangars, warehouses for materials and machinery. The required areas and construction works for the project include the design of the building, necessary drawings for project and the excavation and paving works as shown in Table 18.

Table 18: Construction work

Construction work	Area (m ²)
Administrative offices	40
Labortory	30
Production hall	250
Werehouse	50
Pool	200
Total	570

Machinery and Equipment

It includes the cost of the initial investment, in addition to the processing of the site, the work of plowing , settlement of the land, installation of the system and agricultural seedlings. in addition to

the generator of electricity if the area is not covered by the electric grid to cool the axial, the following table shows the machines and equipment that required for the project:

Table 19: machines and equipment

Machines and equipments	Number	Unit Cost (JD)
Tools and equipment for plowing	1	2500
Field Packets	200	25
irrigation mesh	200	200
Sprayers	1	2000
Wash basin with conveyor sinks	1	2000
Trays (1X2.5 m) of metal mesh or perforated sheet	150	20
Stands for trays	100	20
Grind stainless steel tables (1.7*.8*.8 m)	10	250
Balance and other tools	1	1000
Spare power generators	1	5000
Other (water tanks, pumps, tools, etc.)	1	3000
drying machine or oven	1	4000
Packing machine and deflation machines	1	500
Other Tools	1	2500

Furniture Requirements

Project needs simple furniture for management and permanent employment, that estimated at about 3000 JD, which is simple tables , offices , cabinets and laboratory equipment. The table below shows the details of the furniture needed.

Table 20: Furniture and furnishings required for the project

Furniture	Number	Unit cost (JD)
Office Furniture	1	2000
laboratory equipment	1	1000

Vehicles Requirements

The project needs a pickup to move inside and outside the project as well as a medium transport vehicle for project staff and to be used by the marketing representative to reach the target customers. The following table shows the required transportation:

Table 21: Required Transportation for the Project

Vehicles and transport vehicles	Number	Unit cost (JD)
Medium Transport Vehicle (Pickup)	2	15,000
Tractor	1	12,000

Legal requirements and procedures

Agricultural activity in itself doesn't need a license but as the project seeks to produce a prepared food commodity for domestic and to export to Arab and foreign markets prior to the investment starting, a number of legal and administrative conditions should be met. These include legal facility registration, licensing and approvals and others, and deliver the required documents and payment of the initial administrative costs associated with investing

1. Legal registration, whether it was an individual or company Foundation (solidarity, limited partnership, Special contribution, limited liability company, etc.)
2. Registration of the trade name
3. Professional licensing- a prerequisite for legal establishments to be able to do business.
4. Register at the public institution for social security- Each facility is committed under the provisions of Jordanian law to register in social security
5. Obtain Jordan Institution for Standards and Metrology approval.
6. Identify the specific technical requirements, which the exporters can benefit from the available export opportunities. Most of these requirements can be identified and overcome by how to enter the available EU market at the EU Export Promotion Office. It is important to note that all goods are in conformity with the European Union specifications in order to reach the European market with all careing. Therefore, it protect humans, animals, consumers rights and the environment.

1. Financial Analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

The estimated costs are calculated within the real range of fixed prices for 2017 and are expected to be the same prices for the base year. Project life expectancy is 10 years and the year (0) is the base year in which the investment occurs and the interests start to appear in the first year of the project Year (1) by half of production. The project is fully operational in the second year and continues until the end of the project in the tenth year. The cost of the project is divided into three main sections; investment costs, operational costs and working capital, based on agricultural and engineering information obtained from local experts, tomato producers and field crop specialists.

Calculation Assumptions:

- The weighted average cost of capital used in the study is 11.97% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Income tax on the net project income is 0 % during the life time of the project because its an agriculture project.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Table 22: Annual Growth Rates Assumptions

Annual Growth Rates	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate by (1)	2.00%
Annual Expenses Increase Rate by (2)	2.00%
Annual Increase Rate in Employees Salaries by (3)	3.0%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

Capital Expenditures

The estimated cost of the project is 398,998 JD and it covers land costs, construction works (65,934), machinery and equipment (87,725) , furniture, pre-operating expenses (8,250), working capital (102,401) and vehicle requirement(46,200). In addition 10% will added to contingences, the following table summarize the details of these expenses:

Table 23:Capital expenditure

Capital expenditure	Cost (JD)	Contingency	Cost (JD)	%
Construction works	59,940	10%	65,934	%16.5
Machinery &Equipment	79,750	10%	87,725	%22.0
Furniture	3,000	10%	3,300	%0.8
Vehicles	42,000	10%	46,200	%11.6
Pre-Operating Expenses	7,500	10%	8,250	%2.1
Working capital	170,536	10%	187,589	%47.0
Total (JD)	362,726		398,998	100%

Construction Work

The project needs a number of facilities, most importantly underground wells, a small management building, temporary housing for workers, hangars and warehouses for materials and machinery. Table 24 shows the construction works required for the project :

Table 24:Construction Work Capital Expenditure

Description	Cost (JOD per	Area	Gross Cost/ JD
Administrative offices	120	40	4,800
Labortory	120	30	3,600
Production hall	120	250	30,000
Werehouse	120	50	6,000
Pool	5	200	1,000
Site works and settlement works	2	5,000	10,000

Other Works	-	-	4,540
Total Cost		570	59,940

Machines and equipment

It includes the cost of the initial investment, in addition to the processing of the site, the work of plowing , settlement of the land, Installation of irrigation system, gassel basins, trays, saws, agricultural tractor with complete plows, tomato drying equipment as well as packing machine. in addition to the generator of electricity, the following table shows the machines and equipment that required for the project:

Table 25: Machines and equipment Capital Expenditure

Description	Number	Unit	Gross Cost/ JD
Tools and equipment for plowing	1	2500	2,500
Field Packets	200	25	5,000
irrigation mesh	200	200	40,000
Sprayers	1	2000	2,000
Wash basin with conveyor sinks	1	2000	2,000
Trays (1X2.5 m) of metal mesh or perforated sheet	150	20	3,000
Stands for trays	100	20	2,000
Grind stanless steel tables (1.7*.8*.8 m)	10	250	2,500
Balance and other tools	1	1000	1,000
Spare power generators	1	5000	5,000
Other (water tanks, pumps, tools, etc.)	1	3000	3,000
drying machine or oven	1	4000	4,000
Packing machine and deflation machines	1	500	500
Other Tools	1	-	7,250
Total (JD)			79,750

Furniture Requirements

The table below shows the details of the furniture needed and its costs:

Table 26: Furniture Capital Expenditure

Description	Number	Unit Cost / JD	Gross Cost/
Office Furniture	1	2,000	2,000
laboratory equipment	1	1,000	1,000
Total			3,000

Vehicles Requirements

Table(7) shows the details of the Transporation and Vehicles needed for the project and its costs:

Table 27: Transporation and Vehicles Capex

Transporation and Vehicles	No	Unit Cost/JOD	Total
Medium Transport Vehicle (Pickup)	2	15,000	30,000
Tractor	1	12,000	12,000
Total Cost	3		42,000

Pre Operating Expenses

Consist of a set of expenses that are expended before the start of the project and before any revenue is realized. It consist of Government fees, licenses, marketing expenses, training and transportation expenses which are estimated as follows:

Table 28: Pre-operating expenses

Pre-operating expenses	Estimaited cost (JD)
Governmental fees	1,000
Transportation	1,500
Marketing	3,000
Training	2,000
Total	7,500

Working Capital

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital, which often enough to cover three months :

Table 29:Working Capital

Working Capital	No. of Months	%	Estimated Cost
Operating expenses	6	%50	155,069
General and administrative expenses	3	%25	2,627
Marketing expenses	3	%25	1,440
Indirect Salaries	3	%25	11,400
Total	6	%50	155,069

Depreciations

The following tables illustrate cost of Capex and Depreciation rates:

Table 30:Capex and Depreciation Expenses

Capex Cost and Annual Depreciation Rates	Cost (JOD)	Depreciation Rate	Annual Additions Percentage
Land	0	%0.0	%0.0
Construction works	65,934	%5.0	%0.0
Machinery &Equipment	87,725	%10.0	%2.0
Furniture	3,300	%10.0	%5.0
Vehicles	46,200	%10.0	%0.0
	203,159		

Source of funding

The financing structure of the project consists of loans and Equity. The investment cost of this project is estimated at (398,998) JD, where 50% of the project will be financed through loans (199,499) JD, and the remaining (50%) through (Equity) with (199,499) JD. The following table summarizes the general structure of the required financing:

Table 31: Depreciations and Additions on Construction Works

Capex, Annual Additions and	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total Fixed Assets	203,159	205,079	207,041	209,049	211,102	213,201	215,349	217,546	219,793	222,093
Total Depreciation Expenses	17,019	17,211	17,407	17,608	17,813	18,023	18,238	18,458	18,683	18,913
Total Accumulated Depreciation	17,019	34,230	51,638	69,246	87,059	105,083	123,321	141,779	160,462	179,374
Total Additions	0	1,920	1,963	2,007	2,053	2,100	2,148	2,197	2,248	2,299
Total Net Book Values	186,140	170,848	155,404	139,803	124,042	108,118	92,028	75,767	59,332	42,719

Table 32: Sources of Funding

Funding Sources	Amount	Percentage
Equity	199,499	%50
Loans	199,499	%50
Total	398,998	%100
Use of Fund:	Value	Ratio
Capital Expenditures	203,159	%50.92
Pre-Operating Expenses	8,250	%2.07
Working Capital	187,589	%47.02

Table 33: Loan Assumptions

Loan Assumptions	Measure unit	Assumption Value
Loan ratio	Percentage	50%
Property rights	Percentage	50%
Annual Interest Rate (1)	Percentage	9.50%
Loans Period	Year	5
Loan Starts at year:	Year	2018
Grace Period	Year	0

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Annual Revenues

The sale price per kilogram of the final product estimated (2.0) JD which increased annually by 2%, as indicated in the market study, based on the high price expectations in the exporting and importing countries. The price in Jordan is very exaggerated. With the profit margin for wholesale and wholesale retailers, the sale price is expected to be between 2.5-3 JD per kilogram. In general, when estimating the selling price for financial analysis, the price It must be very competitive, as the product is intended to spread in the local market. As shown in Table (35)

Operating Expenses

Table(36) summarizes the expected operating expenses for the proposed project, starting from agriculture and land working to the raw materials and packaging requirements.

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General and Administrative Expenses

Table (37) summarizes the expected general and administrative expenses for the proposed project (2018-2027), Table (38) shows the marketing expenses, which its about 2% from the revenues based on the study team assumptions.

Human Resources

The total annual salaries and monthly wages has been estimated on 16 months for the purpose of considering annual increments on salaries as well as payments related to social security and health insurance,... etc.

Table (39) summarizes the expected annual salaries and wages, considering annual increment on salaries as per the income statement:

Table 34:Financing loans used

Indicator	2018	2019	2020	2021
beginning of period Balance	199,499	166,495	130,355	90,782
Intrests	18,952	15,817	12,384	8,624
Capital	33,004	36,140	39,573	43,333
Return the loan	51,957	51,957	51,957	51,957
End of period Balance	166,495	130,355	90,782	47,449
The current portion of the loan	199,499	166,495	130,355	90,782
long term loans	18,952	15,817	12,384	8,624

Table 35: Expected Revenues for (2018-2027)

Project Revenues	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total sales (tons)	144	288	288	288	288	288	288	288	288	288
Item Price (JD / ton)	2,000	2,040	2,081	2,122	2,165	2,208	2,252	2,297	2,343	2,390
Total sales (JD)	288,000	587,520	599,270	611,256	623,481	635,951	648,670	661,643	674,876	688,373
Total Project Revenues (JD)	288,000	587,520	599,270	611,256	623,481	635,951	648,670	661,643	674,876	688,373

Table 36: Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Raw Materials and Packaging from Revenues	168,200	168,200	171,564	174,995	178,495	182,065	185,706	189,421	193,209	197,073
Utilities from Revenues	5,760	11,750	11,985	12,225	12,470	12,719	12,973	13,233	13,498	13,767
Maintenance from Revenues	0	588	599	611	623	636	649	662	675	688
Disposables from Revenues	1,440	2,938	2,996	3,056	3,117	3,180	3,243	3,308	3,374	3,442
Insurance expenses from the value of assets	144	294	300	306	312	318	324	331	337	344
Direct Salaries	106,400	109,592	112,880	116,266	119,754	123,347	127,047	130,859	134,784	138,828
Others	28,194	29,336	30,032	30,746	31,477	32,226	32,994	33,781	34,588	35,414

Total	310,138	322,697	330,357	338,206	346,249	354,491	362,938	371,594	380,465	389,557
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Table 37: General and Administrative Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Telecom	500	510	520	531	541	552	563	574	586	598
Hospitality	250	255	260	265	271	276	282	287	293	299
Stationary and Printing	200	204	208	212	216	221	225	230	234	239
Renting	8,000	8,160	8,323	8,490	8,659	8,833	9,009	9,189	9,373	9,561
Consultant	0	0	0	0	0	0	0	0	0	0
Training	500	0	0							
Miscellaneous	955	923	942	961	980	999	1,019	1,040	1,061	1,082
Total	10,507	10,156	10,359	10,566	10,777	10,993	11,213	11,437	11,666	11,899

Table 38: Marketing Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing Expenses	5,760	11,750	11,985	12,225	12,470	12,719	12,973	13,233	13,498	13,767

Table 39: Annual expected salaries and wages (2018-2027)

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect Employees										

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Project Manager / Agricultural Engineer	16,000	16,480	16,974	17,484	18,008	18,548	19,105	19,678	20,268	20,876
Agricultural Engineer	12,800	13,184	13,580	13,987	14,407	14,839	15,284	15,742	16,215	16,701
Food Production Technician	6,400	6,592	6,790	6,993	7,203	7,419	7,642	7,871	8,107	8,351
Marketing and supply representative	5,600	5,768	5,941	6,119	6,303	6,492	6,687	6,887	7,094	7,307
Guard	4,800	4,944	5,092	5,245	5,402	5,565	5,731	5,903	6,080	6,263
Total indirect Salaries	45,600	46,968	48,377	49,828	51,323	52,863	54,449	56,082	57,765	59,498
Direct Employees										
Technical production supervisor	5,600	5,768	5,941	6,119	6,303	6,492	6,687	6,887	7,094	7,307
Workers	96,000	98,880	101,846	104,902	108,049	111,290	114,629	118,068	121,610	125,258
Driver	4,800	4,944	5,092	5,245	5,402	5,565	5,731	5,903	6,080	6,263
Total direct Salaries	106,400	109,592	112,880	116,266	119,754	123,347	127,047	130,859	134,784	138,828
Total	152,000	156,560	161,257	166,095	171,077	176,210	181,496	186,941	192,549	198,326

Income and expenditure statement

Through all financial information and analysis of the project's financial expenses, the estimated income statement for years (2018-2027)

Statement of financial position

Through all financial information and analysis of the project's financial expenses, the estimated income statement for years (2018-2027) as shown in table (41). The statement of financial position is one of the main financial statements on which the project is based, a list showing the financial position of the entity. The list is usually assessed on the last day of the financial year.

Accounts receivable and payables

The preparation of the balance sheet is based on a set of assumptions concerning the timing of the cash flow received and deferred for the subsequent period, with the aim of give every period its revenue and costs. Table (40) shows the assumptions on which the balance sheet was based and the movement of balances of payables during the period of the project (2018-2027)

Table 40:Balance sheet assumptions

Balance sheet assumptions	%
Cash received during the period from accrued income	%83
Cash received during the subsequent period	%17
Cash paid during the period from operating expenses during the fiscal year	%92
Cash paid during the subsequent period	%8
End of period stock	%8

Expected Cashflows Statement

Table (41) shows the expected cash flow statement for the project, where cash flows are shown to be positive for the length of the project period other than the project start year, indicating the project's need for external loans during that period to pay periodic operational surcharges

Table 41: Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	288,000	587,520	599,270	611,256	623,481	635,951	648,670	661,643	674,876	688,373
Gross Operating Revenues	288,000	587,520	599,270	611,256	623,481	635,951	648,670	661,643	674,876	688,373
Operating Expenses	(310,138)	(322,697)	(330,357)	(338,206)	(346,249)	(354,491)	(362,938)	(371,594)	(380,465)	(389,557)
Gross Operating Profit	(22,138)	264,823	268,914	273,050	277,232	281,460	285,732	290,049	294,411	298,816
<i>Gross Profit Percentage</i>	<i>%8-</i>	<i>%45</i>	<i>%45</i>	<i>%45</i>	<i>%44</i>	<i>%44</i>	<i>%44</i>	<i>%44</i>	<i>%44</i>	<i>%43</i>
Salaries and Benefits (Indirect Staff)	(45,600)	(46,968)	(48,377)	(49,828)	(51,323)	(52,863)	(54,449)	(56,082)	(57,765)	(59,498)
General and Administraive Expenses	(10,507)	(10,156)	(10,359)	(10,566)	(10,777)	(10,993)	(11,213)	(11,437)	(11,666)	(11,899)
Markeing Expenses	(5,760)	(11,750)	(11,985)	(12,225)	(12,470)	(12,719)	(12,973)	(13,233)	(13,498)	(13,767)
Pre Operating Expenses	(8,250)									
Gross Indirect Expenses	(70,117)	(68,874)	(70,721)	(72,620)	(74,570)	(76,575)	(78,635)	(80,752)	(82,928)	(85,164)
Income before Interest, Depreciation and Tax	(92,255)	195,948	198,192	200,430	202,662	204,885	207,097	209,297	211,482	213,652
Fixed Assets Depreciations	(17,019)	(17,211)	(17,407)	(17,608)	(17,813)	(18,023)	(18,238)	(18,458)	(18,683)	(18,913)
Income before Tax and Interests	(109,274)	178,737	180,785	182,822	184,848	186,861	188,859	190,839	192,800	194,739

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Bank Interests	(18,952)	(15,817)	(12,384)	(8,624)	(4,508)	0	0	0	0	0
Income Before Tax	(128,227)	162,920	168,401	174,198	180,341	186,861	188,859	190,839	192,800	194,739
Income Tax	0	0	0	0	0	0	0	0	0	0
Net Profit	(128,227)	162,920	168,401	174,198	180,341	186,861	188,859	190,839	192,800	194,739
<i>Net Profit Percentage</i>	<i>%45-</i>	<i>%28</i>	<i>%28</i>	<i>%28</i>	<i>%29</i>	<i>%29</i>	<i>%29</i>	<i>%29</i>	<i>%29</i>	<i>%28</i>
Compulsory Reserves	0	(16,292)	(16,840)	(17,420)	(18,034)	(18,686)	(18,886)	(19,084)	(19,280)	(19,474)
Retained Earnings	(128,227)	18,401	169,962	326,740	489,047	657,222	827,195	998,950	1,172,470	1,347,735

Table 42: Expected Balance Sheet

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Assets										
Current Assets										
Cash	1,439	94,638	237,029	381,580	528,282	729,080	932,006	1,137,046	1,344,184	1,553,400
Inventory	14,017	14,017	14,297	14,583	14,875	15,172	15,476	15,785	16,101	16,423
Accounts Receivable	48,000	97,920	99,878	101,876	103,913	105,992	108,112	110,274	112,479	114,729
Total current Assets	63,456	206,574	351,205	498,039	647,070	850,244	1,055,594	1,263,105	1,472,764	1,684,552
Non Current Assets										
Fixed Assets (net)	186,140	170,848	155,404	139,803	124,042	108,118	92,028	75,767	59,332	42,719

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total Non Current Assets	186,140	170,848	155,404	139,803	124,042	108,118	92,028	75,767	59,332	42,719
Total Assets	249,595	377,422	506,608	637,842	771,112	958,362	1,147,621	1,338,872	1,532,096	1,727,270
Liabilities	-	-	-	-	-					
Current Liabilities										
Remaining amount of Loan	11,828	12,875	13,233	13,601	13,979	14,369	14,769	15,181	15,605	16,040
Total current Liabilities	36,140	39,573	43,333	47,449	0	0	0	0	0	0
Non Current Liabilities	47,968	52,448	56,565	61,050	13,979	14,369	14,769	15,181	15,605	16,040
Long Terms Loans										
Total Long Term Liabilities	130,355	90,782	47,449	0	0	0	0	0	0	0
Total Liabilities	130,355	90,782	47,449	0	0	0	0	0	0	0
Owners Equity	178,323	143,230	104,015	61,050	13,979	14,369	14,769	15,181	15,605	16,040
Shareholders Contributions										
Statutory Reseve	199,499	199,499	199,499	199,499	199,499	199,499	199,499	199,499	199,499	199,499
Retained Profits	0	16,292	33,132	50,552	68,586	87,272	106,158	125,242	144,522	163,996
Total Equity	(128,227)	18,401	169,962	326,740	489,047	657,222	827,195	998,950	1,172,470	1,347,735
Total Liabilities and Equity	71,272	234,193	402,594	576,792	757,132	943,994	1,132,852	1,323,691	1,516,491	1,711,230

Table 43: Expected Cashflows

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash Inflows from Operating Activities										
Net Profit	(128,227)	162,920	168,401	174,198	180,341	186,861	188,859	190,839	192,800	194,739
Bank Interests	18,952	15,817	12,384	8,624	4,508	0	0	0	0	0
Depreciation	17,019	17,211	17,407	17,608	17,813	18,023	18,238	18,458	18,683	18,913
Total Operating Cashflows before	(92,255)	195,948	198,192	200,430	202,662	204,885	207,097	209,297	211,482	213,652
Inventory (Increase/Decrease)	(14,017)	0	(280)	(286)	(292)	(297)	(303)	(310)	(316)	(322)
Accounts Receivables (Increase/Decrease)	(48,000)	(49,920)	(1,958)	(1,998)	(2,038)	(2,078)	(2,120)	(2,162)	(2,205)	(2,250)
Accounts Payables (Increase/Decrease)	11,828	1,047	358	368	379	389	400	412	424	436
Working Capital (Increase/Decrease)	(50,188)	(48,873)	(1,881)	(1,915)	(1,951)	(1,986)	(2,023)	(2,060)	(2,098)	(2,136)
Net Cashflows from Operating Activities	(142,444)	147,075	196,311	198,515	200,711	202,898	205,074	207,237	209,385	211,516
Cashflows from Investments Activities										
Fixed Assets (Procurement)	(203,159)	(1,920)	(1,963)	(2,007)	(2,053)	(2,100)	(2,148)	(2,197)	(2,248)	(2,299)
Net Cashflows from Investments	(203,159)	(1,920)	(1,963)	(2,007)	(2,053)	(2,100)	(2,148)	(2,197)	(2,248)	(2,299)
Cashflows from Financing Activities										
Capital	199,499									

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Loan Amortization	(33,004)	(36,140)	(39,573)	(43,333)	(47,449)	0	0	0	0	0
Bank Interest Rate	(18,952)	(15,817)	(12,384)	(8,624)	(4,508)	0	0	0	0	0
Loans	199,499	0	0	0	0	0	0	0	0	0
Net Cashflows from Financing Activities	347,042	(51,957)	(51,957)	(51,957)	(51,957)	0	0	0	0	0
Net (Increase/Decrease) in Cash	1,439	93,199	142,392	144,551	146,702	200,799	202,926	205,040	207,137	209,216
Cashflows at the Beginning of Period	0	1,439	94,638	237,029	381,580	528,282	729,080	932,006	1,137,046	1,344,184
Cashflows at the End of Period	1,439	94,638	237,029	381,580	528,282	729,080	932,006	1,137,046	1,344,184	1,553,400

Feasibility Indicators

There is more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans, therefore the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Table 44: Risk Premium

Risk Premium	
(1) Risk Free Rate of Return	6.50%
(2) Return to Debt Maturity	7.50%
(3) Market Risk Premium	9.93%
Income Tax Rate	5.00%
(4) Beta	1
Growth Rate	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector,, pages.stern.nyu.edu

Accordingly, the weighted average cost of capital for the project was reached at 12.71% as described in (45).

Table 45: Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%7.50
Market Risk Premium	%9.93
Income Tax Rate	%0
Bets	1.00
Equity	199,499
Loans Value	199,499
Loans	
Cost of Borrowing before Tax	%7.50
Cost of Borrowing After Tax	%7.50

Loans Value	199,499
Loans Percentage	%50
Equity	
Cost of Equity	%16.4
Equity Value	199,499
Equity Percentage	%50
Gross	
Project Value	398,998
Project Percentage	%100
Weighted Average Cost of Capital	%11.97

Payback Period

Table below illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Net Present Value

The net present value of the investment project indicates the difference between the present value of the cash inflows generated by the project and the present value of the project's outflows. If the NPV is positive - ie, the present value of the inflows increase than the cash outflows - the investment project is profitable, and vice versa. If there is more than one investment option, the option that gives the largest net present value is preferred.

Internal Rate of Return

The IRR is one of the most important criteria used for evaluating and comparing different investment projects. It is currently used by the World Bank in all types of financial and economic analysis of projects. As well as by most IFIs when they accept or reject projects submitted for funding. This criterion is at the rate at which the present value of cash flows is equal to the present value of the cash outflows of the project, ie the discount rate at which the net present value of the investment project is zero. The following table shows the results of the financial analysis of the project

Table 46: Financial Analysis Results

Description	Measuring	
Weighted Average Cost of Capital (WACC)	Percentage	%11.97
Net Present Value for Cashflows	JD	1,275,065
Payback Period	Year	3
Internal Rate of Return	Percentage	%33.30

Table 47:List of Net Projected Free Cash Flow for the Years (2018-2027)

	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	Final Value
Income before taxes and interest		(101,024)	178,737	180,785	182,822	184,848	186,861	188,859	190,839	192,800	194,739	
Income tax		0	0	0	0	0	0	0	0	0	0	
Net profit		(101,024)	178,737	180,785	182,822	184,848	186,861	188,859	190,839	192,800	194,739	
Adjustments												
Consumption		17,019	17,211	17,407	17,608	17,813	18,023	18,238	18,458	18,683	18,913	
Change in working capital		(50,188)	(48,873)	(1,881)	(1,915)	(1,951)	(1,986)	(2,023)	(2,060)	(2,098)	(2,136)	
Fixed assets and capital additions		0	(1,920)	(1,963)	(2,007)	(2,053)	(2,100)	(2,148)	(2,197)	(2,248)	(2,299)	
Net Free Cash Flow	-398,998	-134,194	145,155	194,348	196,508	198,658	200,799	202,926	205,040	207,137	209,216	2,731,765

Table 48:Free Net Cash flows Table

Net Cashflows	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	Final Value
Net Free Cashflows	(398,998)	(134,194)	145,155	194,348	196,508	198,658	200,799	202,926	205,040	207,137	209,216	2,731,765
Discount Factor	1.00	0.89	0.80	0.71	0.64	0.57	0.51	0.45	0.40	0.36	0.32	0.32
Net Present Value for Cash Flows	(398,998)	(119,853)	115,789	138,463	125,040	112,900	101,922	91,995	83,020	74,906	67,573	882,308

Table 49: Payback Period

Payback Period	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Free Cashflows	(398,998)	(134,194)	145,155	194,348	196,508	198,658	200,799	202,926	205,040	207,137	209,216
Project Value Returned	398,998	533,192	388,037	193,688	(2,820)	(201,478)	(402,277)	(605,203)	(810,243)	(1,017,380)	(1,226,596)
Payback Period (Year)	3	1	1	1	0	0	0	0	0	0	0

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes the declined in revenues, changes in total investment costs, and increase in operating costs with 10%, which indicate that the project in feasible.

The following table shows that the project will be feasible if the revenues declined with 10% or the operating costs increase by 10%.

Table 50: Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC	The weighted average cost of capital
Original Scenario	33.30%	3	1,275	11.97%
Revenues declined by 10%	24.85%	5	701	11.97%
Operating Expenses Increased by 10%	26.93%	4	879	11.97%
Total Expenses Increased by 10%	25.49%	5	837	11.97%
Increase operating expenses and decrease revenues 10%	16.53%	7	266	11.97%

Conclusions and Recommendations

Based on the profitability analysis and considering that project internal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 1,275,065 Jordan Dinars considering that the project provides 7 Job Opportunities for the governorate residents